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Matthew Docherty

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PROFESSIONAL EXPERIENCE

Oldbaum Services

Stirling, UK

Senior Data Scientist

Jan 2025 - Present

- Oversaw project management, hiring, and personnel development, serving on leadership and recruitment committees.
- Owned company-wide production codebase, managing real-time systems, PR reviews, integration testing and releases.
- Managed small dev team, promoting test-driven Python practices and robust pipeline design.
- Appointed member of 5-person international expert group advising Dutch Government on standards for public data release.
- Built secure software platform, enabling clients to run proprietary code in real-time without accessing underlying logic.
- Scoped £3M+ client campaigns, delivering technical demos and onboarding support.

Data Scientist

Jun 2023 - Jan 2025

- Independently redesigned legacy Python codebase into modular, containerised architecture with automated CI/CD.
- Overhauled data storage strategy by building secure, scalable cloud platform to host high-volume sensitive client data.
- Engineered proprietary ETL pipelines, unlocking 10x storage and 100x speed gains - facilitating company's competitive edge.
- Delivered client-facing technical analyses, certified for reliability and used in mission-critical decision making.
- Conducted solo technical deployments of £5M+ equipment in UK, Japan, South Korea and the Philippines.
- Published original peer-reviewed research, setting new standards for the offshore wind industry.

Trespass

Glasgow, UK

Data Analyst

Jan 2023 - Jun 2023

- Sole data analyst reporting directly to Executive Directors at a global retail brand (£100M+ annual turnover).
- Reconciled £200K+ in sales discrepancies by processing and cleaning 3 years of historical order data using Python workflows.
- Automated daily reporting via Python pipelines, enabling accurate weekend reporting for the first time in company history.
- Led analytics and forecasting strategy to inform high-level decisions for international expansion into Canada and Poland.
- Conducted NLP-driven real-time competitor sentiment analysis to optimise engagement and ROI on £30K email campaign.

ASOS.com

London, UK

NLP Research Intern

Jan 2022 - Sep 2022

- Worked with team of ML researchers on recommender systems for global e-commerce brand (£3B+ annual revenue).
- Designed transformer-based LLMs on sensitive customer data, outperforming internal benchmarks on key engagement KPIs.
- Contributed to the full ML lifecycle, gaining hands-on experience with MLOps workflows for deployment at petabyte-scale.

University College London

London, UK

ML PhD Researcher

Oct 2021 - Nov 2022

- Conducted original research in deep generative Monte-Carlo modelling on big-data.
- Deployed cutting-edge scalable ML pipelines on remote HPC clusters with GPU acceleration.
- Developed DevOps best-practices by maintaining research group's public open-source GitHub repositories (60+ GitHub stars).
- Published peer-reviewed research and presented at top-tier international ML and astrophysics conferences.

EDUCATION

MSci Physics with Astrophysics (1st Class), University of Glasgow

Sep 2016 - Jun 2021

- **Awards:** Masters Thesis Prize (2021), Undergraduate Astronomy Prize (2018), Best Astronomy Lab Student (2017).
- **Scholarships:** RISE Germany (2020), Carnegie Trust (2020), Royal Astronomical Society (2019).

TECHNICAL STACK

- **DevOps & Engineering** - CI/CD (Travis CI, GitHub Actions); Version Control (Git, DVC); Deployment (Docker, Kubernetes, PyPI); HPC (Slurm, CUDA, Apache Spark, Airflow); Cloud (AWS, Azure, GCP); API (REST, SOAP, GCS); Shell (Bash, PowerShell).
- **Data** - Cloud (Snowflake, Azure Blob, AWS S3 Bucket); SQL (MySQL, SQLServer, PostgreSQL); Excel (VBA, Power Query).
- **Python** - Analysis (Pandas, Dask, PySpark); Visualisation (Streamlit, Plotly-Dash, Seaborn, Matplotlib); Dev (FastAPI, Django).
- **Machine Learning** - Frameworks (TensorFlow, PyTorch, Scikit-Learn, JAX); NLP (SpaCy, NLTK, Gensim).

SELECTED PUBLICATIONS

- A. Oldroyd, M. Young, **M. M. Docherty**, et al. "Comparison of scanning lidar hard-targetting methods for offshore wind." (2024).
- M. A. Price, M. Mars, **M. M. Docherty**, et al. "Fast emulation of anisotropies induced in the CMB by cosmic strings." (2023).
- A. S. Mancini, **M. M. Docherty**, M. A. Price, and J. D. McEwen. "Bayesian model comparison for simulation-based inference." (2023).